Probability & statistics

Home Assignment - 3

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2) (a) $P(X=3)$

WKT sum of probabilities $= 1$

$0.1 + 0.3 + 0.3 + x + 0.1 = 1$

$x + 0.8 = 1$

$\boxed{x = 0.2}$ $\boxed{P(X=3) = 0.2}$

(b) $P(X \geq 2)$

$P(X=2) + P(X=3) + P(X=4)$

$0.3 + 0.2 + 0.1$

$\boxed{P(X \geq 2) = 0.6}$

5) given $n = 20$

$P(\text{defective}) = 10\% = \frac{10}{100} = 0.1$

$q \Rightarrow 1 - P \Rightarrow 0.9$

(a) $P(X=0)$

$\Rightarrow {}^nC_r \, p^r \, q^{1-r}$

${}^{20}C_0 \, (0.1)^0 \, (0.9)^{20}$

$1 \times 1 \times (0.9)^{20}$ $\Rightarrow \boxed{P(X=0) = 0.12}$

(b) $P(X=2)$

${}^{20}C_2 \, (0.1)^2 \, (0.9)^{18}$

$\frac{20 \times 19}{2} \, (0.1)^2 \, (0.9)^{18}$

$\boxed{P(X=2) = 0.28}$

(c) $P(X=2) + P(X=3)$

$0.28 + {}^{20}C_3 \, (0.1)^3 \, (0.9)^{17} \Rightarrow 0.28 + \frac{\overset{10}{20} \times 19 \times \overset{6}{18}}{3 \times 2} \, (0.1)^3 \, (0.9)^{17}$

$\boxed{P(X=2) + P(X=3) = 0.47}$

6) $p = \frac{25}{100} = 0.25$

$q = 0.75$

$n = 9$ $\quad P(x=0) + P(x=1) + P(x=2) + P(x=3) = P(x<4)$

$\Rightarrow {}^9C_0 (0.25)^0 (0.75)^9 + {}^9C_1 (0.25)^1 (0.75)^8 + {}^9C_2 (0.25)^2 (0.75)^7$

$+ {}^9C_3 (0.25)^3 (0.75)^6$

$\Rightarrow (0.75)^9 + 9 \times (0.25) \times (0.75)^8 + \frac{9 \times 8}{2} \times (0.25)^2 (0.75)^7$

$+ \frac{9 \times 8 \times 7}{3 \times 2} \times (0.25)^3 (0.75)^6$

$\boxed{P(x<4) \Rightarrow 0.834}$

7) given $p = 0.4$

$q = 0.6$

$n = 15$

ⓐ mean, variance

$\boxed{\text{mean} = np} \Rightarrow 15 \times 0.4 \Rightarrow 6$

$\boxed{\text{variance} = npq} \Rightarrow 15 \times 0.4 \times 0.6 \Rightarrow 3.6$

ⓑ $P(x=5)$

${}^{15}C_5 (0.4)^5 (0.6)^{10} \Rightarrow \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2} \times (0.4)^5 (0.6)^{10}$

$\boxed{P(x=5) = 0.18}$

ⓒ $P(X \geq 10)$

$P(x=10) + P(x=11) + P(x=12) + P(x=13) + P(x=14) + P(x=15)$

${}^{15}C_{10} (0.4)^{10} (0.6)^5 + {}^{15}C_{11} (0.4)^{11} (0.6)^4 + {}^{15}C_{12} (0.4)^{12} (0.6)^3 + {}^{15}C_{13} (0.4)^{13} (0.6)^2$

$+ {}^{15}C_{14} (0.4)^{14} \times (0.6)^1 + {}^{15}C_{15} (0.4)^{15} (0.6)^0$

$\Rightarrow \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2} (0.4)^{10} (0.6)^5 + \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2} (0.4)^{11} (0.6)^4 + \frac{15 \times 14 \times 13}{3 \times 2 \times 1} (0.4)^{12} (0.6)^3$

$+ \frac{15 \times 14}{2 \times 1} (0.4)^{13} (0.6)^2 + 15 (0.4)^{14} \times (0.6) + (1) (0.4)^{15}$

$P(x \geq 10) = 0.0338$

(d) $P(3 \le x \le 8)$

$P(x=3) + P(x=4) + P(x=5) + P(x=6) + P(x=7) + P(x=8)$

$^{15}C_3 (0.4)^3 (0.6)^{12} + {}^{15}C_4 (0.4)^4 (0.6)^{11} + {}^{15}C_5 (0.4)^5 (0.6)^{10} + {}^{15}C_6 (0.4)^6 (0.6)^9$

$+ {}^{15}C_7 (0.4)^7 (0.6)^8 + {}^{15}C_8 (0.4)^8 (0.6)^7$

$\Rightarrow \frac{15 \times 14 \times 13}{3 \times 2} (0.4)^3 (0.6)^{12} + \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} (0.4)^4 (0.6)^{11} + \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1} (0.4)^5 (0.6)^{10}$

$+ \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10}{6 \times 5 \times 4 \times 3 \times 2} (0.4)^6 (0.6)^9 + \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9}{7 \times 6 \times 5 \times 4 \times 3 \times 2} (0.4)^7 (0.6)^8$

$+ \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} (0.4)^8 (0.6)^7$

$\Rightarrow 0.8804$

(e) $P(x \le 3)$

$P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$\Rightarrow {}^{15}C_0 (0.4)^0 (0.6)^{15} + {}^{15}C_1 (0.4)^1 (0.6)^{14} + {}^{15}C_2 (0.4)^2 (0.6)^{13}$

$+ {}^{15}C_3 (0.4)^3 (0.6)^{12}$

$\Rightarrow 0.095$

(f) $P(x=0)$

$= {}^{15}C_0 (0.4)^0 (0.6)^{15} = 0.000047$

1) given

$$f(x) = \begin{cases} Kx^2 & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

(i) $\boxed{\text{For a valid PDF } \int_{-\infty}^{\infty} f(x)\,dx = 1}$

$\int_0^2 Kx^2 = 1 \Rightarrow K \left[\frac{x^3}{3}\right]_0^2 = 1 \Rightarrow K[8-0] = 3$

$\Rightarrow \boxed{K = 3/8}$

(ii) $\int_0^1 kx^2 \Rightarrow \frac{3}{8}\int_0^1 x^2 \Rightarrow \frac{\cancel{3}}{8\times\cancel{3}}\left[x^3\right]_0^1 \Rightarrow \frac{1}{8} = 0.0125$

3) given $f(x) = K/_{2^x}$ $\quad x = 0, 1, 2, 3, 4$

a) $\sum_{x=0}^{4} f(x) = 1$

$$K\left(\frac{0}{2^0} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}\right) = 1$$

$$\boxed{K = \frac{16}{31}}$$

(b) cumulative distribution function

$x = 0 \Rightarrow \frac{16}{31 \times 2^0} = \frac{16}{31}$

$x = 1 \Rightarrow \frac{\cancel{16}8}{31 \times \cancel{2}} = \frac{8}{31}$

$x = 2 \Rightarrow \frac{\cancel{16}4}{31 \times \cancel{4}} = \frac{4}{31}$

$x = 3 \Rightarrow \frac{16}{31 \times 8} = \frac{2}{31}$

$x = 4 \Rightarrow \frac{16}{31 \times 16} = \frac{1}{31}$

CDF

$x < 0$

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{16}{31} & 0 \le x < 1 \\ \frac{24}{31} & 1 \le x < 2 \\ \frac{28}{31} & 2 \le x < 3 \\ \frac{30}{31} & 3 \le x < 4 \\ 1 & x \ge 4 \end{cases}$$

4) $f(x) = \begin{cases} k\sqrt{x} & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$

ⓐ $\boxed{\text{For a valid PDF } \int_{-\infty}^{\infty} f(x)\,dx = 1}$

$$\int_0^1 k\sqrt{x} = 1 \quad k\int_0^1 \sqrt{x} \Rightarrow k\int_0^1 x^{1/2} \Rightarrow$$

$$k\left[\frac{x^{1/2+1}}{\frac{1}{2}+1}\right]_0^1 = 1 \Rightarrow \frac{2}{3}k\left[x^{3/2}\right]_0^1 = 1$$

$$\boxed{k = 3/2}$$

ⓑ $\int_{0.3}^{0.6} k\sqrt{x} \Rightarrow \frac{3}{2}\int_{0.3}^{0.6} x^{1/2} \Rightarrow \frac{3}{2}\cdot\frac{2}{3}\left[x^{3/2}\right]_{0.3}^{0.6}$

$= (1) \times (0.6)^{3/2} - (0.3)^{3/2}$

$\Rightarrow 0.3$

8) ⓐ Two cards drawn from 52 cards

$P(\text{two diamonds}) = \frac{13}{52} \times \frac{12}{51} = \frac{1}{17}$

Number of trials $n = 51$

$\lambda = np = 51 \times \frac{1}{17} = 3$

Poisson formula:-

$$\boxed{P(x=n) = \frac{e^{-\lambda}\lambda^x}{x!}}$$

$P(x \geq 3) = 1 - [P(0) + P(1) + P(2)]$

$P(0) = e^{-3} = 0.0498$

$P(1) = 3e^{-3} = 0.1494$

$P(2) = \frac{3^2}{2}e^{-3} = 0.224$

$P(x \geq 3) = 1 - (P(0) + P(1) + P(2)) = 0.577$

(b) $p = 0.02$, $n = 100$

$\lambda = np = 100 \times 0.02 = 2$

(i) At least one defective

$P(x \geq 1) = 1 - P(0)$

$P(0) = e^{-2} = 0.1353$

$P(x \geq 1) = 1 - 0.1353$

$P(x \geq 1) = 0.8647$

(ii) exactly 7 defective

$$P(7) = \frac{e^{-2} \cdot 2^7}{7!}$$

$P(7) = 0.0034$

(iii) $P(1 < x < 8)$

$= P(2) + P(3) + P(4) + P(5) + P(6) + P(7) = 0.593$

4) Mean $\mu = 2040$

Standard deviation $\sigma = 60$

Total bulbs $N = 2000$

$$z = \frac{x - \mu}{\sigma}$$

(a) more than 2150 hours

$$z = \frac{2150 - 2040}{60} = 1.83$$

from z-table:

$P(z > 1.83) = 0.03$

Number of bulbs:

$2000 \times 0.0336 = 67.2$

(b) less than 1950 hours

$$z = \frac{1950 - 2040}{60} = -1.5$$

$P(z < -1.5) = 0.06$

$2000 \times 0.06 = 133.6 \Rightarrow 134$ bulbs

(c) more than 1920 but less than 2160

$$z_1 = \frac{1920 - 2040}{60} = -2$$

$$z_2 = \frac{2160 - 2040}{60} = 2$$

$P(-2 < z < 2) = 0.95$

$2000 \times 0.9544 = 1908.8 = 1909$ bulbs

(2) exactly 1960 hours

$P(x = 1960) = 0$

10) $P(x < 40) = 0.10 \rightarrow z = -1.28$

$P(x > 75) = 0.05 \rightarrow z = 1.645$

$$z = \frac{x - \mu}{\sigma}$$

(1)

$$\frac{40 - \mu}{\sigma} = -1.28$$

$$\frac{75 - \mu}{\sigma} = -1.645$$

$\sigma = 11.97, \ \mu = 55.32$

(2)

$$z_1 = \frac{50 - 55.32}{11.97} = -0.44$$

$$z_2 = \frac{60 - 55.32}{11.97} = 0.39$$

$P(z < 0.39) = 0.6517$

$P(z < -0.44) = 0.33$

$P(50 < x < 60) = 0.6517 - 0.33$

$= 0.3217$

32.17% students in second division.